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3. INTRODUCTION

The EduGear Resource Management System is a web-based application developed using PHP and MySQL. It was designed to help campus support staff manage and track academic equipment such as laptops, projectors, cameras, microphones, and other devices.

The system replaces manual record keeping with a digital platform that allows staff to manage equipment, issue loans to students, process returns, and generate daily summaries.

4. SYSTEM OVERVIEW

The system is used by campus employees to manage equipment efficiently. It allows users to:

- Log into the system securely
- Add, update, view, and delete equipment
- Issue equipment to students
- Process returned items
- View daily loan statistics

The system ensures that equipment availability is always updated in real time.

5. SYSTEM DESIGN (OOP CONCEPTS)

Although the system is built using PHP procedural and object-based logic, object-oriented programming concepts are applied in key areas.

5.1 Classes Used

The system can be structured into the following logical classes:

- Employee Class
- Equipment Class
- Loan Class
- Database Connection Class

5.2 Constructors

Constructors are used to initialise objects when data is passed into the system, such as employee details or equipment information.

5.3 Encapsulation

Encapsulation is demonstrated by keeping data secure and only allowing controlled access through methods. For example, database queries are only executed through prepared statements.

5.4 Polymorphism / Logical Variation

Different equipment categories (Laptop, Projector, Camera, etc.) behave differently in terms of usage but are managed using the same system structure.

5.5 Error Handling

The system uses basic error handling through validation checks and database error prevention using prepared statements.

6. DATABASE DESIGN

The system uses a MySQL database named `edugear_db`.

Tables:

Employees

- `employee_id` (Primary Key)
- `name`
- `surname`
- `employee_number`
- `campus_name`

Equipment

- `equipment_id` (Primary Key)
- `equipment_name`
- `equipment_category`
- `equipment_condition`
- `availability_status`
- `replacement_value`

Loans

- `loan_id` (Primary Key)
- `student_name`
- `student_surname`
- `student_number`
- `equipment_id`
- `loan_date`
- `expected_return_date`
- `return_date`
- `loan_status`

7. SYSTEM FEATURES

7.1 Login System

Employees log in using their name and employee number. A session is created to restrict access to authorised users only.

7.2 Equipment Management

Users can:

- Add new equipment
- View equipment list
- Update equipment details
- Delete equipment

7.3 Loan Management

Staff can issue equipment to students. The system records:

- Student details
- Equipment borrowed
- Loan date
- Expected return date
- Loan status

Unavailable equipment cannot be issued.

7.4 Return Processing

When equipment is returned:

- Loan status is updated to “Returned”
- Equipment becomes available again
- Return date is recorded

7.5 Daily Summary

The system displays:

- Total loans issued today
- Total returns today
- Active loans
- Overdue loans
- Total replacement value of active loans

8. SECURITY AND VALIDATION

The system includes basic security measures:

- Session-based login protection
- Prepared statements to prevent SQL injection

- Required field validation
- Numeric validation for values like replacement cost
- Regex validation for student numbers
- User-friendly error messages

9. USER GUIDE

Step 1: Login

Enter employee name and employee number to access the system.

Step 2: Dashboard

Navigate through the menu to access different system features.

Step 3: Manage Equipment

Add, update, or delete equipment items.

Step 4: Issue Equipment

Select available equipment and assign it to a student.

Step 5: Return Equipment

Mark equipment as returned when it is brought back.

Step 6: View Summary

Check daily statistics of system activity.

10. ERROR HANDLING

The system handles errors in the following ways:

- Empty fields are not allowed
- Invalid data formats are rejected
- Database errors are managed using prepared statements
- Users receive clear error messages when something goes wrong

11. SCREENSHOTS (YOU MUST ADD)

Include screenshots of:

- Login page
- Dashboard
- Add equipment page

- View equipment table
- Loan system
- Return system
- Summary page
- Database structure

12. CONCLUSION

The EduGear system successfully demonstrates a functional equipment loan management system built using PHP and MySQL. It allows efficient tracking of equipment and ensures proper management of loans and returns. The system improves accuracy, reduces manual work, and provides a structured digital solution for campus support operations.